Here is a comprehensive guide breaking down every problem statement. This document is designed to help you visualize the "Build" phase for each topic so you can match it to your team's skills.

🚀 Hackathon Problem Statement Guide

Goal: Select the problem that best matches your tech stack (Web, App, AI, or IoT) and interest.

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1. Agritech Solutions (Agriculture)

Focus: Helping farmers using Data, IoT, or Simple Apps.

PS Code	The "Real" Problem	Solution Idea (What to Build)	Difficulty
AS-1	Farmers plant crops blindly without knowing market demand, leading to price crashes.	**Smart Crop Planner:** A web dashboard where a farmer enters their location/soil type, and the system suggests the *most profitable* crop based on current market prices and weather forecasts.	⭐⭐⭐ (Data heavy)
AS-2	Crops get destroyed by pests or disease before harvest because farmers don't notice early signs.	**Doctor Crop App:** An image-based mobile app where a farmer takes a photo of a leaf, and AI (TensorFlow) detects the disease and suggests a cure.	⭐⭐⭐⭐ (AI/Image)

AS-3	Villages waste water due to poor planning.	**Village Water Dashboard:** An IoT or simulation dashboard that tracks water levels in village tanks and suggests a schedule: "Field A gets water on Monday, Field B on Tuesday."	⭐⭐⭐ (IoT/Logic)
AS-4	Gov schemes exist, but farmers don't know how to apply or find them.	**"Kisan Sahay" Portal:** A super-simple app (vernacular language) that matches a farmer's profile (acres, crop, income) to eligible government schemes and auto-fills forms.	⭐⭐ (Good for Web Devs)
AS-5	Farming advice is expensive or generic.	**Agri-Chatbot:** A WhatsApp-integrated AI bot that gives free, specific advice like "How much fertilizer for cotton in Pune?" without needing a human expert.	⭐⭐ (GenAI/Chatbot)

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2. Traffic Safety

Focus: Saving lives, reducing accidents, and managing chaos.

PS Code	The "Real"	Solution Idea	Difficulty

	Problem	(What to Build)	
TS-1	Authorities don't know *where* accidents happen most frequently.	**Accident Heatmap:** A data visualization tool (using Mapbox) that plots historical accident data to show "Black Spots" so police can place barricades there.	⭐⭐ (Data Viz)
TS-2	Ambulances reach late because no one reports accidents instantly.	**Auto-SOS System:** An App/IoT prototype that uses phone acceleration sensors (Accelerometer) to detect a crash and automatically sends GPS coordinates to nearby hospitals.	⭐⭐⭐ (Mobile Sensors)
TS-3	People drive rashly because there is no incentive to drive safely.	**Safe Driver Gamification:** An app that tracks speed and braking. Safe driving earns "Karma Points" redeemable for fuel coupons or insurance discounts.	⭐⭐⭐ (Mobile Tracking)
TS-4	**(Recommended)** Last-mile connectivity issues (Home to Metro).	**MetroLink Aggregator:** *See detailed plan in previous response.* A shared-ride booking system for autos/cabs from	⭐⭐⭐ (Full Stack)

		home to transit hubs.	
TS-5	Traffic signals run on fixed timers even when the road is empty.	**Smart Signals:** Use camera feeds (Computer Vision) to count cars. If Lane A is empty and Lane B is full, the AI turns Lane B Green immediately.	⭐⭐⭐⭐⭐ (Advanced AI)

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3. Environment

Focus: Sustainability, Pollution, and Green Habits.

PS Code	The "Real" Problem	Solution Idea (What to Build)	Difficulty
EN-1	Environmental data (AQI, Carbon footprint) is too complex for normal people.	**Eco-Scorecard:** A simplified widget/app that tells a user "Your air quality is equal to smoking 2 cigarettes today" (making data relatable).	⭐⭐ (UI/UX Focus)
EN-2	People mix wet and dry waste, making recycling impossible.	**Smart Bin:** An IoT bin with a camera that scans trash. If you throw plastic in the "Wet" bin, it beeps/alerts you to correct it.	⭐⭐⭐⭐ (IoT + AI)
EN-3	No motivation to be eco-friendly.	**Green Rewards Platform:** A	⭐⭐⭐ (Social/Web3)

		community app where users upload photos of planting trees or cleaning trash to earn crypto-tokens or local store discounts.	
EN-4	Air pollution is hard to pinpoint to a specific source.	**Pollution Hunter:** A drone or sensor network dashboard that identifies *exactly* which factory or area is releasing smoke in real-time.	⭐⭐⭐⭐ (Hardware/IoT)
EN-5	Cities react to pollution *after* it happens, not before.	**Pollution Predictor:** An AI model that analyzes wind + traffic + construction data to predict "Smog likely tomorrow" and warns authorities.	⭐⭐⭐⭐ (Data Science)

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4. Disaster Management

Focus: Emergency response and coordination.

PS Code	The "Real" Problem	Solution Idea (What to Build)	Difficulty
DM-1	Internet fails during disasters; alerts don't reach people.	**Offline Alert Mesh:** An app that uses "Mesh	⭐⭐⭐⭐⭐ (Complex Networking)

		Networking" (Bluetooth/WiFi Direct) to pass messages from phone to phone without internet.	
DM-2	Authorities don't know which buildings will collapse in an earthquake.	**Risk Audit Tool:** A checklist app for civil engineers to inspect buildings and upload data, generating a "Risk Map" for the city.	⭐⭐ (CRUD App)
DM-3	**(Recommended)** Chaos during rescue; volunteers don't know where to go.	**Relief Commander:** A centralized dashboard connecting Victims (Need help), Volunteers (Have boat/food), and Gov (Coordination). Like "Uber for Rescue."	⭐⭐⭐ (Full Stack)
DM-4	Assessing damage (broken bridges, houses) takes weeks.	**Drone Survey Analysis:** A system where users upload photos/videos of damage, and AI estimates the repair cost automatically.	⭐⭐⭐⭐ (AI/Image)
DM-5	People die because warnings come too late (e.g., floods).	**Early Warning Siren:** An IoT water-level sensor connected to a cloud system that	⭐⭐⭐ (IoT)

		triggers SMS/Siren alerts when water crosses a danger mark.	

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5. Edutech (Education)

Focus: Learning, Exams, and Careers.

PS Code	The "Real" Problem	Solution Idea (What to Build)	Difficulty
ET-1	Every student learns at a different pace, but classes are "one size fits all."	**Adaptive Learning Bot:** A tutor platform where if a student fails a quiz, the system suggests *easier* videos. If they pass, it gives *harder* challenges.	⭐⭐⭐ (Logic/Algo)
ET-2	Students choose wrong careers due to peer pressure.	**Career Compass:** A psychometric test platform + AI that analyzes a student's marks and interests to suggest: "You are 90% match for Data Science."	⭐⭐ (Logic/Data)
ET-3	Education apps don't work in villages with 2G internet.	**Lite-Learn:** A "Progressive Web App" (PWA) that downloads lessons when internet is fast and lets students learn	⭐⭐⭐ (Technical Optimization)

		offline later.	
ET-4	Exam cheating and lost papers.	**Transparent Exam Vault:** A blockchain-based or secure encrypted system where answer sheets are scanned and stored so they can't be tampered with.	⭐⭐⭐⭐ (Security)
ET-5	Exams test memory, not skill.	**Skill Simulator:** Instead of multiple choice questions, a game-like test where a coding student actually fixes a bug, or a mechanic virtually fixes an engine.	⭐⭐⭐⭐ (Gamification)

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6. Healthcare

Focus: Prevention, Access, and Mental Health.

PS Code	The "Real" Problem	Solution Idea (What to Build)	Difficulty
HC-1	People ignore symptoms until it's too late.	**Symptom Checker:** An AI chat interface where you describe "stomach pain + fever" and it predicts "Possible Typhoid - Visit Doctor."	⭐⭐⭐ (GenAI)

HC-2	Hospital bills are shocking/hidden.	**Hospital Compare:** A "Yelp/Zomato" for hospitals where patients rate treatment costs and transparency, helping others choose.	⭐⭐ (Web App)
HC-3	Remote patients need constant monitoring.	**Remote Vitals:** An app that reads heart rate using the phone camera (photoplethysmography) and sends daily reports to the doctor.	⭐⭐⭐⭐ (Advanced Tech)
HC-4	Rural nurses lack expert knowledge.	**Nurse Assistant:** A tablet app for rural workers that guides them: "If patient has high BP, check X, Y, Z" (Decision Tree logic).	⭐⭐ (Logic/App)
HC-5	Mental health is stigmatized and therapy is expensive.	**Mood Buddy:** An anonymous, friendly AI chatbot for students to vent stress, offering CBT (Cognitive Behavioral Therapy) exercises.	⭐⭐ (GenAI)

How to Choose? (My Final Advice)

1. **If you are a Full Stack Developer (MERN/Python):**
   - **Best:** DM-3 (Rescue Platform) or AS-4 (Farmer Schemes).
   - *Why:* These require strong database, user management, and dashboard skills. They

are "complete" products.

2. **If you are an AI/ML Enthusiast:**
   - **Best:** TS-5 (Traffic Signals) or AS-2 (Crop Disease).
   - *Why:* These have a clear "AI Model" core (Computer Vision) that impresses judges visually.
3. **If you are an App Developer (Flutter/React Native):**
   - **Best:** TS-4 (Last Mile) or TS-2 (SOS App).
   - *Why:* These rely heavily on GPS, Maps, and smooth mobile UX.
4. **If you want to solve a unique/creative problem:**
   - **Best:** ET-3 (Low Network Education).
   - *Why:* Solving for "Low Internet" is a very hard technical challenge that engineering judges love.